MZUZU UNIVERSITY

FACULTY OF INFORMATION SCIENCE AND COMMUNICATIONS

DEPARTMENT OF INFORMATION AND COMMUNICATION TECHNOLOGY

BACHELOR OF SCIENCE IN INFORMATION AND COMMUNICATION TECHNOLOGY (GENERIC)

END OF SEMESTER EXAMINATION

ICT 4702: MODELING AND SIMULATION

TIME ALLOWED: 3 HOURS **DATE: 15TH FEBRUARY, 2018**

INSTRUCTIONS

i. This paper contains **TWO** pages. Please check.
ii. There are **FOURTEEN QUESTIONS** in this paper. Please check.
iii. Answer **ALL** questions.
iv. Marks for each question are indicated.

1. Define the following terms:
   a. Neighborhood function. **[1 Mark]**
   b. Pseudo-random number. **[1 Mark]**
   c. Simulation. **[1 Mark]**
   d. Event. **[1 Mark]**
   e. Attribute. **[1 Mark]**

2. Describe any **five** applications of Cellular automata. **[10 Marks]**

   a. Explain the difference between stochastic and deterministic models **[5 Marks]**

3. Compare the modeling process with the scientific method. **[10 Marks]**

4. The National Bank of Malawi has 12 tellers with an exponentially distributed service rate with a mean of 10 minutes. Customers arrive at the bank with exponentially distributed inter arrival times with an average inter arrival time of 2 minutes. There is a single queue for customers to wait in, but the queue is restricted to 40 customers.

   a. What observation can you make about the departure rate of customers from the bank? **[5 Marks]**

   b. What is the probability that a customer shows up at the bank and is immediately served? **[5 Marks]**

5. You are using a simulation to study the steady-state average waiting time of customers in a complex system. However, you find that, because of the simulation complexity, the computer time needed to arrive at a steady-state value is excessive, leading to poor estimates of the steady-state values. How might you improve the speed at which the simulation converges? **[10 Marks]**

6. Customers calls arrive at a software support center with exponentially distributed inter arrival times with an average time between arrivals of 15 minutes. What is the probability that 5 customers will arrive between 1 p.m. and 2:30 p.m.? **[5 Marks]**

7. Mzuzu University's ICT department has 5 printers in the computer room for the use of the department's 1500 undergraduate and 2000 graduate students. Print jobs arrive uniformly during the day and have an exponentially distributed page count. They are spooled on a single hard disk while waiting for the first printer to be available. In the standard notation of queuing systems, how would this queuing system be characterized? **[6 Marks]**

8. Using the multiplicative congruential method, find the period of the generator for $a=13$, $m=2^6=64$ and $X_0=1,2$ *and* 3 **[9 Marks]**

9. With an aid of the formula, explain how Combined Linear Congruential method of random number generation works. **[6 Marks]**

10. Identify one area where simulation is applicable and explain why simulation is important by giving a relevant example. **[5 Marks]**

11. "You don't test a system by stopping it". Explain this famous quote with reference to simulation. **[5 Marks]**

12. Consider an M/M/1 queuing system with an arrival rate $\lambda=0.4$ and service rate$\mu=0.5$.
    a. Compute the system load and tell if the system stable or not? **[2 Marks]**
    b. Compute the average number of customers in the system. **[2 Marks]**

13. Consider a computer system with two processors and one infinite queue. Tasks arrive to the processors according to a Poisson process with a rate $\lambda=20$ tasks/sec and each server has a processing speed of C=2 Mbit/sec. The service time is exponentially distributed with a mean $1/\mu=50$ ms.

    a. What is the service rate of each processor expressed in tasks/sec. **[5 Marks]**
    b. Compute the average task size (L) in Kbits. **[5 Marks]**

14. Why is probability theory of interest to performance modeling of information systems? **[5 Marks]**

END OF QUESTION PAPER